Practical 9

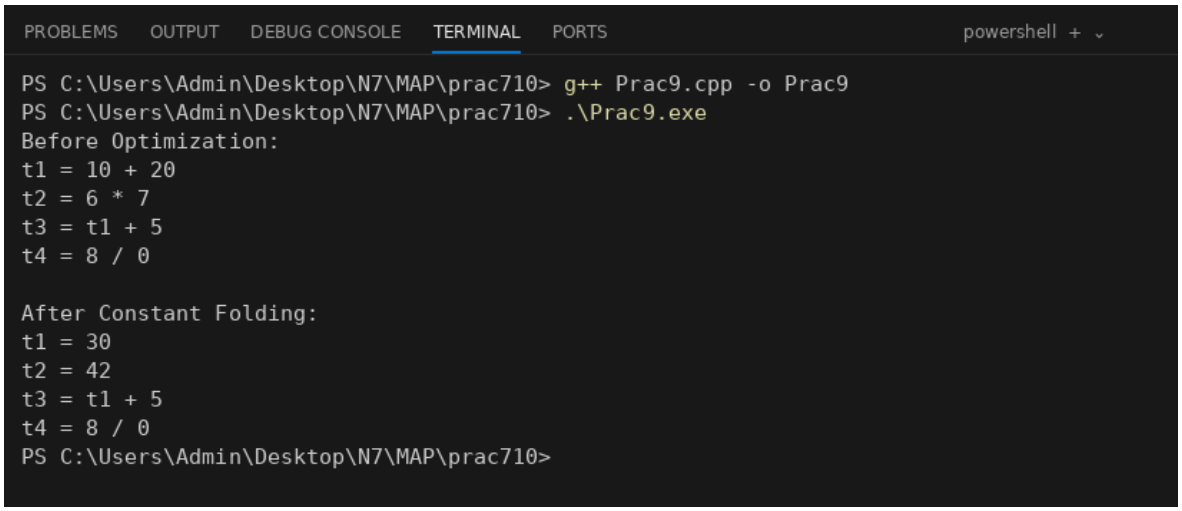
Aim: Implement Code Optimization Techniques: Implement any one code optimization technique to improve the efficiency of the generated intermediate code. (Use C/C++ language.)

Prac9.cpp:

```
#include <iostream>
#include <string>
#include <cctype>
using namespace std;

bool isNumber(string s)
{
    if (s.empty())
        return false;
    for (int i = 0; i < s.size(); i++)
        if (!isdigit(s[i]))
            return false;
    return true;
}

int main()
{
    string x[] = {"t1", "t2", "t3", "t4"};
    string a[] = {"10", "6", "t1", "8"};
    string op[] = {"+", "*", "+", "/"};
    string b[] = {"20", "7", "5", "0"};

    cout << "Before Optimization:" << endl;
    for (int i = 0; i < 4; i++)
        cout << x[i] << " = " << a[i] << " " << op[i] << " " << b[i] << endl;

    cout << "\nAfter Constant Folding:" << endl;
    for (int i = 0; i < 4; i++)
    {
        if (isNumber(a[i]) && isNumber(b[i]) && !(op[i] == "/" && b[i] == "0"))
        {
            int p = stoi(a[i]), q = stoi(b[i]), r;
            if (op[i] == "+") r = p + q;
            if (op[i] == "-") r = p - q;
            if (op[i] == "*") r = p * q;
            if (op[i] == "/") r = p / q;
            cout << x[i] << " = " << r << endl;
        }
        else
            cout << x[i] << " = " << a[i] << " " << op[i] << " " << b[i] << endl;
    }
    return 0;
}
```

Output:

Statements with both operands as constants are evaluated at compile time.



t3 = t1 + 5 stays unchanged because t1 is not a constant, and t4 = 8 / 0 stays unchanged because division by zero is unsafe.